



- 1 (a)** Use standard results from the list of formulae (MF19) to find $\sum_{r=1}^n (8r^3 + 12r^2 + 4r + 3)$ in terms of n , simplifying your answer. [3]

[illegible]



and hence use the method of differences to find $\sum_{r=1}^n \frac{4r^3 + 6r^2 + 4r + 1}{r^4(r+1)^4}$. [5]

This image shows a full page of a notebook or worksheet template. It consists of approximately 20 evenly spaced horizontal dotted lines across the entire width of the page, providing a guide for handwriting practice. The background is plain white, and there are no margins, text, or other markings present.

(c) Deduce the value of $\sum_{r=1}^{\infty} \frac{4r^3 + 6r^2 + 4r + 1}{r^4(r+1)^4}$. [1]
